

How Do Undergraduate Students Make Decisions When Adopting Generative Artificial Intelligence for Academic Purposes?

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Abstract

Generative artificial intelligence (GAI) has received increased attention in higher education. While recent work on AI literacy has developed AI literacy curricula to guide student use of GAI, researchers have called for empirical studies to investigate students' understandings of GAI tools. This ongoing study explores factors that shape students' decisions on when and how to adopt GAI in academic contexts, including cognitive, contextual, organizational, and demographic factors, through a survey and follow-up interviews. Survey results showed 1) perceived usefulness of GAI based on improving grades and achieving learning goals predicts adopting intention, 2) academic conditions, gender, ethnicity, school year, and location of secondary education mediate GAI perceptions. Preliminary interview results suggested that additional factors, including time pressure, resource availability, and institutional work culture, shape student interaction with GAI tools. These findings contribute to AI literacy research by providing insights into student GAI adoption and implications for designing AI literacy curricula.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Generative artificial intelligence, AI literacy, AI adoption, higher education, mixed-method, student perception

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1 Introduction

Generative artificial intelligence (GAI) in higher education has received increased attention in recent years. It has been described as revolutionary and transformative, challenging the traditional role of higher education institutions due to its capability to assist in students' knowledge and skill acquisition [9, 21]. However, research on the impact of GAI use on students' learning in higher education has yielded mixed results. For instance, a meta-analysis found that overall, the use of GAI on postsecondary levels resulted in improved

learning performance, interest, and self-efficacy [35], while others found that the use of GAI tools resulted in homogeneous essays and less brain neural connectivity [17], and reported concerns such as inaccuracy and over-reliance on GAI tools [37].

The double-edged nature of GAI highlights the importance of AI literacy, defined as a set of competencies that allow users to critically evaluate and collaborate with GAI technologies [19]. Existing studies have highlighted important aspects of AI literacy education, such as knowledge of GAI applications, ability to use and critically evaluate GAI, and ethical considerations of GAI technologies [23]. Other studies developed AI literacy curricula for higher education [16, 30]. However, researchers have also pointed out a lack of consensus on the core AI competencies that should be included in AI literacy curricula, calling for research that empirically explores students' understanding of these AI competencies [18, 19].

Some studies have drawn from technology acceptance frameworks to study students' perceptions of GAI tools and factors that shape GAI adoption in academic contexts. For instance, studies have found that perceived usefulness, effort expectancy, and technology self-efficacy predict students' intention to adopt GAI tools [1, 4, 14, 26]. While these studies provided insights into the cognitive factors that shape the perception and adoption of GAI tools, most studies conceptualized adoption of GAI as a binary variable, rather than a series of decisions which are also shaped by contextual and organizational factors, such as time pressure, type of assignments, and institutional work culture. This limits the transferability of results in supporting the development of AI literacy curricula, which focus on educating students when and how to engage with GAI technologies responsibly.

The present study aims to provide a picture of how students make decisions when engaging with GAI tools in different instances, and support the design of AI literacy curriculum through understanding contextual, organizational, and demographic factors that shape students' decisions to engage with GAI tools in academic contexts. My research question is: How do cognitive, contextual, organizational, and demographic factors shape the ways undergraduate students 1) decide to adopt GAI, and 2) determine when and how to interact with it for academic purposes?

2 Related Work

2.1 Theoretical Models of Technology Adoption

The Diffusion of Innovation Theory (DOI) explores the process by which innovations spread among communities over time [27]. DOI proposes that five characteristics of innovations shape adoption: *relative advantage* over existing technologies, *compatibility* to user values, *complexity* of use, *trialability*, and *observability* of outcomes [27]. While DOI explains technology adoption on societal levels,



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the Technology Acceptance Model (TAM) and subsequent models, such as Unified Theory of Acceptance and Use of Technology (UTAUT), explore how perception shapes technology acceptance on an individual level. TAM proposes that perceived ease of use and perceived usefulness of technological products shape behavioral intention, which predicts adoption [7, 33]. Subsequent models propose additional factors that shape behavioral intention, including performance expectancy, effort expectancy, social influence, and facilitating conditions, all of which are mediated by individual factors such as gender, age, and experience [34].

2.2 AI Literacy

Prior work has developed different conceptualizations of AI literacy. Ng and colleagues [23] proposed four aspects of AI literacy, including understanding, using, evaluating and creating, and knowing ethical issues of AI. Long and Magerko [19] proposed 13 core AI competencies, like recognizing AI-powered artifacts, understanding the strengths and weaknesses of AI, and imagining future AI.

Most studies on AI literacy education focused on K-12 contexts, and those conducted in higher education contexts have focused on developing AI literacy curricula [12, 18]. A study developed and tested an AI literacy curriculum on postsecondary students across academic backgrounds and found improvements in students' understanding and awareness of ethical concerns regarding GAI technologies [16]. Existing work has also taken different approaches to developing AI literacy curriculum, such as drawing from K-12 approaches to develop curriculum for adults [15] or emphasizing aspects like experiential learning [30]. Another study proposed an AI literacy framework based on responsibilities of professional and social roles, such as executive managers, policymakers, or citizens [36]. While many curricula have been developed, researchers have also noted a lack of consensus in the core competencies that underlie AI literacy [18].

2.3 Students' Perception and Adoption of GAI

Prior work has explored factors that shape students' adoption of GAI tools. Many utilized different versions of TAM or UTAUT, and found that factors including perceived usefulness [14, 26], perceived ease of use [14], subjective norm [26], performance expectancy [1, 4], social influence [4], and effort expectancy [1] predict adoption of GAI tools. Another study approached GAI tool adoption with Theory of Planned Behavior and found that perceived strengths of GAI tools predict positive attitudes, which shape students' intention to adopt.

Other empirical work explored postsecondary students' experiences and perceptions of GAI technologies in higher education contexts, focusing on aspects including familiarity, attitudes, usage, perceived benefits and concerns, and the role of demographic factors [2, 3, 5, 8, 13, 20, 24, 25, 29, 31, 32, 37]. Studies found that students demonstrated familiarity of GAI tools [25, 29, 32] and expressed a range of perceived benefits and concerns on GAI use for academic purposes. Benefits include improved academic performance [8, 28], personalized learning support [5, 24, 28, 37], saving time [24]; while concerns include accuracy [5, 8, 25], privacy [5], academic integrity [20, 37], over-reliance on GAI tools [37], ethical considerations, and potential impact on personal development

and career prospects [5]. Researchers also found mixed effects of demographic factors, such as gender and field of study, on student perceptions [8, 20, 31].

3 Methods

The study used a mixed-methods sequential design with an exploratory survey followed by interviews. The survey explored cognitive and contextual factors that shape adoption decisions, while interviews provide explanations to survey results and insights into additional contextual and organizational factors that shape students' adoption decisions. The survey concluded in October 2025, and interviews are still in progress as of January 2026. The study received exemption from the Institutional Review Board at Grinnell College.

3.1 Background

The study was conducted at Grinnell College, a small, private, undergraduate liberal arts college in the Midwestern United States. At Grinnell, students are encouraged to take classes across disciplines regardless of their major(s), which allows us to examine the role of academic conditions as most students have considerable experience across academic disciplines.

3.2 Survey

3.2.1 Design. The online survey had two sections and was created via Qualtrics. The first section collected demographic information, including school year, gender, ethnicity, major(s), and world region of secondary education. The second section used a within-participant design to examine the role of academic conditions, including Humanities (*HUM*), Social Sciences (*SS*), and Science (*SC*). Each condition consisted of two assignment scenarios representing different types of writing assignments typical of the academic division at the institution. The assignment types for each academic condition were literary analysis and reflection paper (*HUM*), literature synthesis and quantitative research paper (*SS*), and statistical report and guided research summary (*SC*), respectively. Participants were randomly assigned to one of the two assignments for each academic condition, and the order of the conditions was counterbalanced. All scenarios were created under consultation with writing instructors and students majoring in respective academic divisions.

For each assignment scenario, participants were presented with the following three prompts, as well as eight specific GAI adoption methods, and asked to evaluate each method in the context of the assignment on a scale from one to five.

- (1) How likely are you to use GAI in the following ways to assist completion of the assignment?
- (2) How useful do you consider using GAI in the following ways to obtain good grades for the assignment?
- (3) How useful do you consider using GAI in the following ways to achieve learning goals of the assignment?

Participants were given all eight specific ways of adopting GAI and asked to rate their likelihood of using GAI (*BL*, behavioral likelihood), and perceived usefulness based on attaining better grades (*PU_{grades}*) and perceived usefulness based on achieving learning goals (*PU_{learn}*), on a scale from 1 (not at all/not useful) to 5 (very likely/very useful). I selected adoption methods based on prior work

Table 1: Participant Demographics ($N = 149$)

Demographic	n (%)
Year	
Senior	52 (35%)
Junior	31 (21%)
Sophomore	32 (21%)
Freshmen	34 (23%)
Gender	
Female	90 (60%)
Male	45 (30%)
Non-binary	14 (9.4%)
Ethnicity	
Asian	31 (21%)
Black/African American	2 (1.3%)
Hispanic/Latinx	5 (3.4%)
Other/Two or more races	19 (13%)
White/Caucasian	92 (62%)

[6, 10, 11, 22] and consultation with the writing center staff. The final list of adoption methods include 1) *exploratory research*, 2) *search for information or examples*, 3) *explain theories or concepts*, 4) *summarize readings*, 5) *generate writing outline*, 6) *generate paragraphs for submission*, 7) *provide feedback to completed or near-completed work*, and 8) *edit grammar, spelling, or punctuation*. Participants were given the opportunity to express interest in a follow-up interview at the end of the survey.

3.2.2 Participants. All participants were current Grinnell students recruited via emails or the psychology department research participation program. The survey received 164 valid responses, out of which 149 were included in the analysis. Table 1 shows the demographic makeup of survey participants.

3.2.3 Model & Data Analysis. I modeled the data with a mixed-effects ordinal logistic regression model using R package *ordinal*¹, of which response values are seen as ordered ranks instead of continuous numeric values. Individual random effects accounted for baseline attitudinal and dispositional differences for GAI by allowing varying intercepts for each individual. As the variance in participant perception on the Likert scale questions may have violated the proportional odds assumption, I focused on the association between variables and differential trends with less emphasis on exact coefficient values. Preliminary analysis showed that assignment scenarios within each academic condition did not significantly impact participants' perceptions. Hence, responses within each academic condition were merged for analysis.

Initial models failed to converge due to a small sample size. To stabilize the model without sacrificing important control variables, I 1) removed 15 observations that showed no variance across all questions, adoption methods, and conditions in the dataset, and 2) refactored Likert scale response values from five to three levels due to sparsity in larger response values. This ensured model convergence and stable parameter estimation.

¹<https://cran.r-project.org/web/packages/ordinal/index.html>

3.3 Follow-up Interviews

I selected interviewees from survey participants who indicated interest in follow-up interviews, based on purposeful sampling principles that maximize diversity in demographic characteristics and perceptions of GAI. I recorded interviews upon participant consent, and transcribed recordings with Whisper, a downloadable speech recognition model developed by OpenAI². The transcripts were then adjusted by the researcher, and will be analyzed via thematic analysis once data collection is complete. Interview questions are included in Appendix A.

4 Findings

4.1 Survey

4.1.1 Behavioral Intention and Perceived Usefulness. Overall, controlling for adoption method, academic condition, and individual baseline differences, PU_{grades} and PU_{learn} were significant positive predictors of BL ($OR_{grades} = 2.28, p < 0.001$; $OR_{learn} = 1.65, p < 0.001$). Examination of adoption-method-level differences showed patterns consistent with overall results. However, while PU_{grades} and PU_{learn} were both significant predictors of BL ($p < 0.001$), estimated parameter strengths differed across adoption methods. The 95% confidence intervals of PU_{grades} and PU_{learn} coefficients had significant overlaps for six out of eight adoption methods. Therefore, I was unable to conclude whether the strength of the relationship between PU_{grades} and PU_{learn} differed for those adoption methods. Nonetheless, I found that PU_{learn} had a stronger association with behavioral likelihood for *exploratory research* than PU_{grade} , and the opposite trend was observed for *summarizing readings*. In conclusion, while overall, PU_{grades} and PU_{learn} are significant positive predictors of BL, adoption-method-level differences suggested that motivation for adopting GAI may differ across adoption methods.

4.1.2 Academic Condition. Results suggested that after controlling for questions, items, and individual baseline differences, participants' perceptions significantly differed across the academic divisions of the assignment. Compared to the reference condition (SS), participants reported lower values for HUM and SC conditions ($OR_{SC} = 0.88, p < 0.05$; $OR_{HUM} = 0.70, p < 0.001$). The large standard deviation of individual random effect control ($SD = 1.92$) indicated substantial baseline differences across participants' evaluations of BL, PU_{grade} , and PU_{learn} .

However, adoption-method-level models suggested that after controlling for PU_{grade} , PU_{learn} , and individual baseline differences, academic condition only appeared to predict behavioral intention for some of the adoption methods, including *exploratory research* ($OR_{SC} = 0.81, p < 0.001$; $OR_{HUM} = 0.48, p < 0.001$), *search info/examples* ($OR_{SC} = 0.69, p < 0.001$; $OR_{HUM} = 0.46, p < 0.001$), and *generate outline* ($OR_{SC} = 0.56, p < 0.001$; $OR_{HUM} = 0.53, p < 0.001$). For the three methods, participants reported less intention to use GAI for HUM and SC assignments compared to SS assignments. Thus, results suggested that academic conditions mediate student GAI use only for specific adoption methods.

²<https://github.com/openai/whisper>

4.1.3 Demographic Characteristics. Results suggested that school year, location of secondary school, gender, and ethnicity mediated perceptions of GAI tools. All models controlled for question, academic condition, adoption method, and individual baseline differences to examine the role of demographic variables³.

Results showed that both school year ($OR = 0.75, p < 0.05$) and location of secondary education ($OR = -1.73, p < 0.001$) were significant predictors of positive perceptions, suggesting that underclassmen and those who completed secondary education outside of the U.S. tend to report more positive perceptions of GAI compared to their counterparts. Perceptions of academic GAI use also significantly differed across gender and ethnicity⁴ lines. Compared to male students (reference group), female students reported less preferable perceptions ($OR = 0.71, p < 0.001$) while non-binary students showed no significant differences from male students ($p = 0.12$). Compared to White/Caucasian students (reference group), Asian students reported more preferable attitudes ($OR = 3.78, p < 0.001$), while students in the Other category showed no significant difference from the reference group ($p = 0.94$).

4.2 Preliminary Interview Results

Preliminary interview results revealed additional contextual and organizational factors that shape students' GAI adoption. For example, a participant believed that students should generally refrain from using GAI for academic purposes, but also mentioned using it once as she "was in a time crunch" and "needed to get that done." Other participants cited tight deadlines and the unavailability of teaching assistants as the main reasons for using GAI, showing how contextual factors shape students' decisions to adopt GAI tools.

Participants also discussed how organizational factors shape GAI adoption through creating incentives. For instance, participants described academics at Grinnell as "really rigorous and hard to keep up", and believed that the lack of "a sustainable culture of work where students are able and equipped to do the work that they're being assigned to" makes students inclined to turn to GAI.

The results also contextualized observed survey trends. For example, a participant believed that GAI tools were more helpful in the social sciences as the field is "very evidence driven, so when you're looking for sources it's very good at pulling those up", and less helpful in humanities as coursework "require you to have a lot of original thought", showing how disciplinary differences mediate the way students interact with GAI.

5 Discussion

The survey results were consistent with prior findings that perceived usefulness predicts students' intention to adopt GAI tools [14, 26], and demographic factors, such as gender, have mediating effects on GAI perceptions [8, 31]. Interviews contextualized survey results by elucidating the driving mechanisms of disciplinary variances and revealing additional contextual and organizational factors, such as time pressure and institutional work culture, that shape students' interaction with GAI. For instance, the case of a

participant using GAI under time pressure, despite the belief that students should refrain from its use, demonstrated how interviews capture nuances that quantitative surveys often overlook.

The present study will bring empirical insights to inform AI literacy efforts in higher education upon completion. Firstly, moving beyond the prevailing operationalization of adoption as a binary outcome, the study views GAI adoption as a series of choices that students make based on their dispositions and contextual factors, allowing for a more comprehensive understanding of student decision-making with GAI. Secondly, comparing student perceptions on GAI use across disciplines helps distinguish between students' disciplinary dispositions (i.e., computer science versus English majors) and their responses to disciplinary expectations (i.e., coding assignments versus literary analysis). This helps us understand whether AI literacy curricula should be rooted in specific disciplines rather than generic approaches. Next, by studying the factors that shape students' interactions with GAI, the study aims to identify critical themes of AI literacy education to facilitate informed and responsible use of GAI tools. Lastly, the study also reveals how institutions can adjust their resources to better support students' learning and reduce GAI reliance in academic contexts.

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³Due to the small sample size, I was unable to create question-level models. Hence, all models in this section consider three questions altogether as participant attitude towards academic GAI use.

⁴I merged categories with < 20 observations into Other, resulting in three ethnicity categories: White/Caucasian, Asian, and Other.

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A Interview Questions

- (1) Please tell me about yourself, such as school year, major, and concentrations, where you’re from, or any other background information that will help me understand your perspective on generative AI.
- (2) Have you used generative AI in any academic capacities, such as assignments, readings, or exam preparation? And if yes, can you briefly give me an overview of how you use it?
- (3) Can you elaborate on the frequency and/or scenarios that make you more likely to use generative AI?
- (4) (If the participant has experience using generative AI) How useful did you find it? In what ways did you find it helpful?
- (5) Have you taken classes from different academic disciplines? If yes, do you find generative AI useful in different ways for coursework from different disciplines?
- (6) In general, what would you consider as appropriate usage of generative AI for academic purposes and why?
- (7) On the flip side, what would you consider as inappropriate usage of generative AI for academic purposes and why?
- (8) Is there anything else that you would like to add about the use of generative AI at Grinnell or in higher education?